NAAC A+

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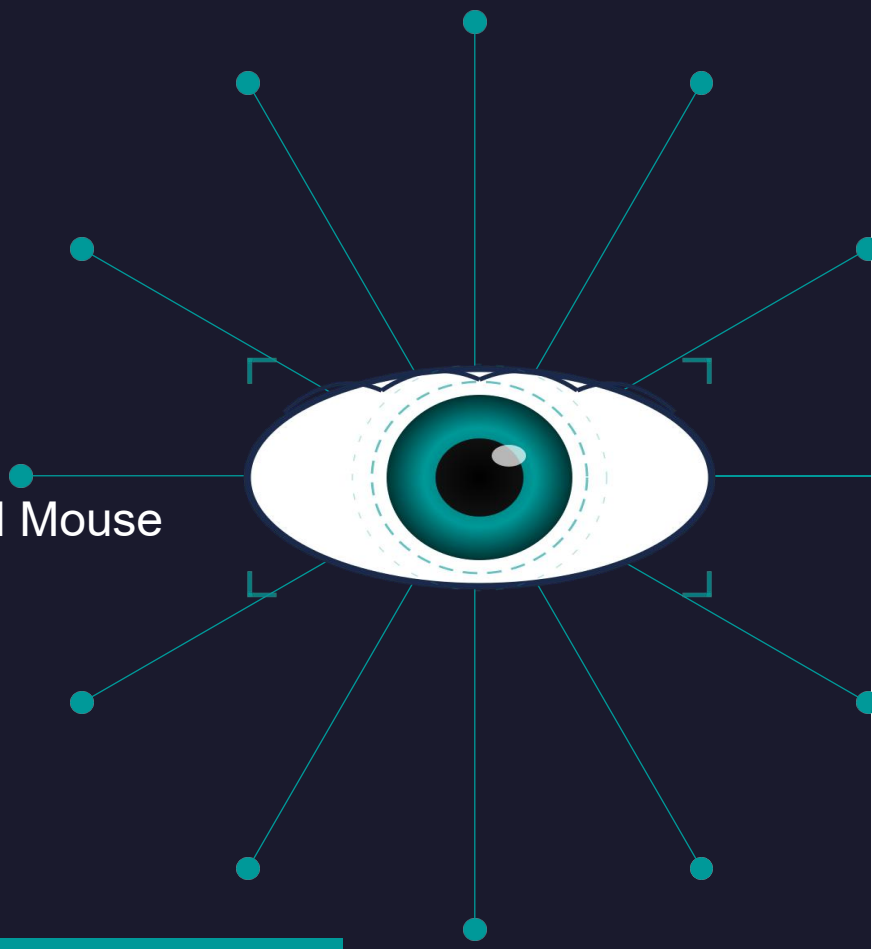
NEURO - CURSOR

Multipurpose Eye-Controlled Mouse



Imagine controlling your computer — clicking, scrolling, navigating — using only your eyes.
No hands. No extra hardware. Just a webcam and a face.
Welcome to Neuro-Cursor: a real-time, software-only gaze interface that replaces the conventional mouse through eye movements, winks, and subtle head motion.

Zero ₹	48	<60 px
Hardware Cost	Landmarks/Fr	Gaze Accuracy

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WHAT IS NEURO-CURSOR?

Neuro-Cursor is a fully software-based, real-time eye-tracking mouse replacement built on a standard webcam. No additional hardware is needed — it runs on any laptop with a built-in camera. The system uses MediaPipe FaceMesh to detect 478 facial landmarks per frame, including dedicated iris refinement points, to track iris positions and estimate gaze direction at real-time frame rates.

A conventional mouse requires fine motor control of the hand and wrist. Neuro-Cursor, by contrast, requires only the natural movement of the eyes. Look at a target, wink to click, glance at screen edges to scroll — and a subtle head turn provides coarse cursor positioning for rapid navigation across the full screen.

Neuro-Cursor uses MediaPipe FaceMesh with refine_landmarks=True to extract pupil positions from landmark indices 468–477. A degree-2 polynomial regression model, fitted with RANSAC over a 16-point + smooth-pursuit calibration, maps these gaze coordinates to screen positions. An Adaptive Kalman filter removes involuntary jitter, and PyAutoGUI drives OS-level cursor movement and all click actions.

The system's gesture vocabulary — winks for clicks, bilateral blinks for double-click, and edge-gaze for scrolling — provides all standard mouse functions without any physical limb movement, entirely through deliberate and detectable eye gestures.

HOW DOES IT DIFFER?

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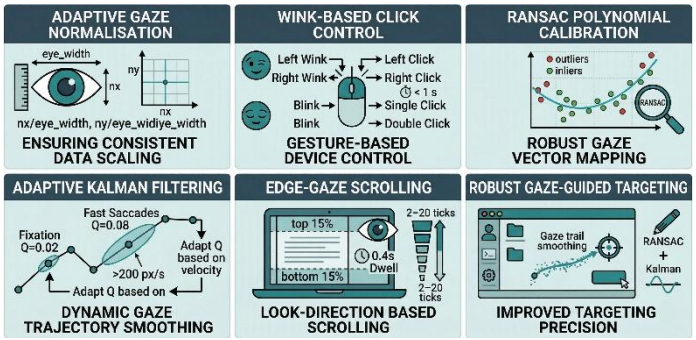
FEATURE	TRADITIONAL SYSTEMS	ADVANCED EYE-TRACKING SYSTEM
Face Detection	Basic face/eye detection	MediaPipe Face Mesh (468 landmarks) for high precision
Calibration	Simple or manual calibration	16-point polynomial calibration with RANSAC outlier rejection
Interaction	Mouse / keyboard required	Hands-free control using eye + head movement
Click Mechanism	Physical mouse clicks	Wink-based clicks (left, right, double blink)
Cursor Stability	Jittery or raw tracking	Adaptive Kalman + Median + Deadzone filtering
Scrolling	Scroll wheel / touchpad	Edge-gaze scrolling with dwell timer
Movement Control	Direct pointer movement	Head movement assist (nose tracking + deadzone)

MediaPipe FaceMesh with refine_landmarks=True activates the iris sub-model, appending landmark indices 468–477 — ten points that precisely annotate both iris contours.
Left iris centre is extracted from index 473; right from 468.
This sub-pixel iris localisation is the foundation of Neuro-Cursor's gaze

KEY CAPABILITIES



■ Adaptive Gaze Normalisation

nx and ny both divided by eye_width — a critical fix ensuring consistent numeric scale on both axes, which is essential for polynomial regression accuracy.

■ Wink-Based Click Control

Left wink = left click. Right wink = right click. Bilateral blink = single click. Double blink within 1 second = double click.

■ RANSAC Polynomial Calibration

Degree-2 polynomial fitted with RANSAC (20 iterations, 80 px threshold) maps 2D gaze vectors to 2D screen coordinates robustly.

■ Adaptive Kalman Filtering

4-state filter (x, vx, y, vy) with velocity-adaptive process noise: Q=0.02 at fixation, Q=0.08 during fast saccades (>200 px/s).

■ Edge-Gaze Scrolling

Looking at top or bottom 15% of screen triggers proportional scroll (2–20 ticks) after a 0.4 s dwell period.

THE NUMBERS

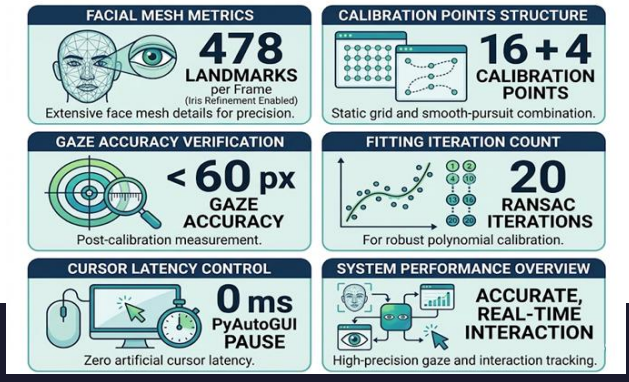
478 MediaPipe landmarks per frame (iris refinement enabled)

16 + 4 Calibration points: static grid + smooth-pursuit paths

< 60 px Target gaze accuracy on post-calibration verification

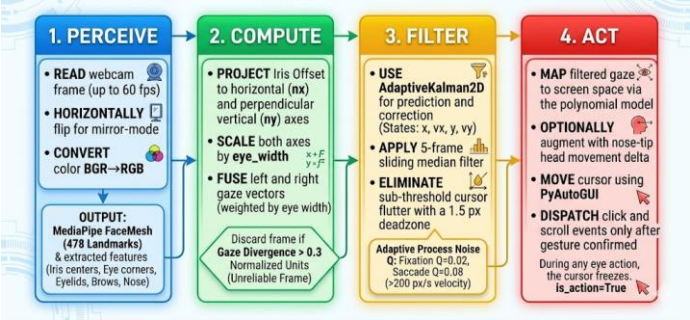
20 RANSAC iterations for polynomial calibration fitting

0 ms PyAutoGUI pause — zero artificial cursor latency



THE BLUEPRINT: HOW NEURO-CURSOR WORKS

Neuro-Cursor processes every webcam frame through a six-stage pipeline before issuing a cursor command to the operating system. The pipeline mirrors a perception–computation–action loop, implemented entirely in real-time computer vision using Python on commodity hardware.



1. PERCEIVE

Each frame is read from the webcam at up to 60 fps, horizontally flipped for mirror-mode interaction, and converted BGR→RGB. MediaPipe FaceMesh returns 478 normalised (x,y,z) landmarks. Iris centres (indices 468 and 473), eye corner pairs (33/133 left, 362/263 right), vertical eyelid pairs (159/145 left, 386/374 right), brow points, and nose tip (index 1) are extracted.

2. COMPUTE

The iris offset is projected onto each eye's horizontal axis (nx) and perpendicular vertical axis (ny), both divided by eye_width for consistent scale. Left and right gaze vectors are fused weighted by each eye's width. Divergence >0.3 normalised units flags an unreliable frame and discards it.

3. FILTER

AdaptiveKalman2D (states: x, vx, y, vy) predicts and corrects each position. Process noise Q switches between 0.02 (fixation) and 0.08 (saccade, velocity >200 px/s). A 5-frame sliding median then removes residual spikes. A 1.5 px deadzone eliminates sub-threshold cursor flutter.

4. ACT

Filtered gaze is mapped to screen space via the polynomial model, optionally augmented by nose-tip head movement delta. PyAutoGUI moves the cursor. During any eye action (is_action=True), the cursor freezes. Click and scroll events are dispatched only after the gesture state machine resolves a confirmed wink or blink.

KEY COMPONENTS

- **MediaPipe FaceMesh: Google's 478-point 3D face model with iris refinement (refine_landmarks=True) for sub-pixel iris centre detection.**
- **AdaptiveKalman2D: 4-state constant-velocity Kalman filter with velocity-adaptive process noise (Q=0.02/0.08, R=15.0).**
- **Polynomial Regression + RANSAC: Degree-2 model [gx, gy, gx^2, gy^2, gx·gy, 1] fitted on inlier calibration points for gaze-to-screen mapping.**
- **Eye Aspect Ratio (EAR): ||top–bottom|| / (||left–right|| + 1e-6). Thresholds: WINK_CLOSED=0.19, WINK_OPEN=0.25, BLINK_BOTH=0.22. Per-eye tracking enables wink disambiguation.**
- **IQR Outlier Rejection: 1.5× IQR filtering on calibration gaze samples before median computation.**
- **CLAHE Enhancement: L channel in LAB colour space enhanced (clipLimit=2.0, 4×4 grid) for eye ROI display.**

GESTURE VOCABULARY

Left Eye Wink

EAR_L < 0.19 AND EAR_R > 0.25, duration 0.10–0.80 s → Left Mouse Click

Right Eye Wink

EAR_R < 0.19 AND EAR_L > 0.25, duration 0.10–0.80 s → Right Mouse Click

Double Bilateral Blink

Two blinks (both EAR < 0.22) within 1.0 s → Double Click

Edge-Gaze Scrolling

Cursor in top or bottom 15% of screen after 0.4 s dwell → Proportional scroll 2–20 ticks

The cursor is completely frozen (is_action=True) whenever any eye is closed — whether for a wink or blink. This prevents the cursor from drifting to an unintended target while a click gesture is in progress. The frozen position is held until the eye reopens and the gesture state machine resolves the action.

REAL-WORLD EXAMPLE

Consider a user with ALS who has lost reliable arm function. With Neuro-Cursor: they look at a browser icon — the cursor moves there. A left wink opens it. They look at the address bar and left-wink to click, then type via an on-screen keyboard app. They scroll a webpage by gazing at the bottom edge. They right-click a file by winking the right eye. All of this on a standard laptop, with zero additional hardware purchases and zero rupees spent on assistive devices.

FROM THEORY TO PRACTICE: NEURO-CURSOR AT WORK

Neuro-Cursor is not just a demonstration. Across accessibility, healthcare, and human–computer interaction research, this technology addresses real, unsolved problems for users who cannot reliably operate a conventional mouse.

■ 1. ASSISTIVE TECHNOLOGY

The primary application is as a hands-free pointing device for users with ALS, cerebral palsy, spinal cord injuries, or upper-limb motor impairments. Since only a webcam is required, deployment is possible in homes, hospitals, and rehab centres without any hardware procurement. The 16-point + smooth-pursuit calibration adapts to each user's unique eye anatomy and webcam placement.

■ 2. TOUCHLESS KIOSK CONTROL

Contactless public interfaces have seen strong demand. Neuro-Cursor can drive information kiosks, ATMs, or medical terminals where physical touch is undesirable. Edge-gaze scroll and wink-click provide all interaction needed for menu navigation without any touch, eliminating surface contamination risk entirely.

■ 3. STERILE ENVIRONMENT OPERATION

Surgeons reviewing scans or notes during a procedure cannot touch a keyboard without breaking sterility. Neuro-Cursor enables hands-free navigation of imaging software using only gaze and winks, with no additional hardware that would need to be sterilised or draped before each procedure.

■ 4. GAMING & IMMERSIVE EXPERIENCE

Head movement assist and gaze-driven cursor augment game controllers — the eye controls aim or camera direction while hands manage other inputs. The dual-input model (gaze + head movement from nose-tip tracking) increases degrees of freedom available without any additional peripherals or hardware investment.

■ 5. HCI RESEARCH PLATFORM

The open-source Python codebase — with its modular Kalman filter, pluggable polynomial calibration, and clean gesture state machine — is a research-ready platform for studying gaze-based HCI, testing new calibration strategies, and benchmarking alternative landmark detection models against MediaPipe FaceMesh.

PERFORMANCE AT A GLANCE

< 60 px EXCELLENT gaze accuracy on 5-point post-calibration verification

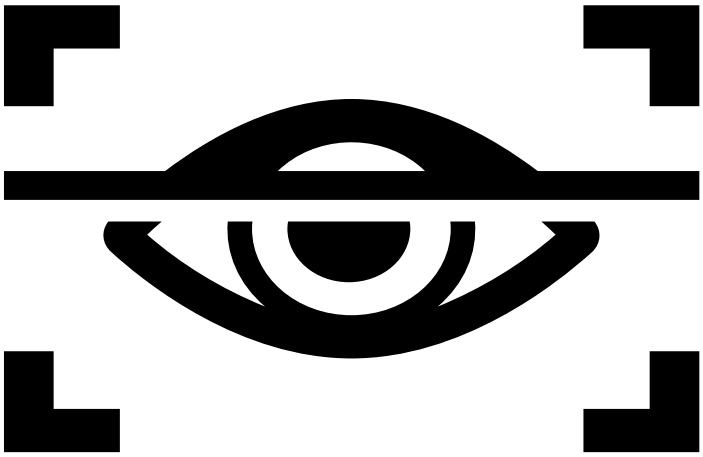
1.5 px Deadzone radius suppressing sub-threshold cursor flutter

0 ms PyAutoGUI pause — zero artificial latency in cursor movement

5-frame Median window reducing residual Kalman filter jitter

0.40 s Edge-gaze scroll dwell time before activation

The 16-point static grid + 4-path smooth-pursuit Phase 2 calibration gives comprehensive screen coverage including edge and corner regions that static grids miss. RANSAC polynomial fitting then selects only geometrically consistent samples, discarding noisy outliers, before the least-squares model is fitted on the final inlier set.



CALIBRATION PIPELINE

Neuro-Cursor uses a two-phase calibration designed to achieve maximum gaze-to-screen accuracy across the full display, including edges and corners that single-phase grids typically miss.

Phase 1 - 16-Point Static Grid:

Fullscreen 4×4 dot grid. Each point: 1.0 s stabilisation + 30 gaze samples. IQR filtering (1.5× IQR) + consistency check (std < 0.025). Median sample stored as calibration pair. Nose position averaged to set head-movement reference.

Phase 2 — Smooth Pursuit:

Dot moves centre→corner (4 paths, 3 s each). Samples from middle 80% of each path added at every 5th frame, providing edge-coverage data that static-only grids miss.

RANSAC Fitting:

20 iterations — each fits a degree-2 polynomial on 6 random points, then counts inliers (error < 80 px). Best inlier set used for final least-squares fit of coefficient vectors Cx and Cy.

Verification:

5 test points displayed. Median gaze error rated: EXCELLENT (<60 px), GOOD (<100 px), FAIR (<150 px), POOR (≥50 px). User can accept calibration or trigger a full recalibration session.

CHALLENGES AHEAD

■ Lighting Sensitivity: Accuracy degrades under strong backlight or very low ambient light. Adaptive histogram equalisation on the full frame before landmark detection is a planned mitigation.

■ Glasses Occlusion: Thick-framed glasses can partially block iris landmarks 468–477. The system falls back to coarser contour estimates, but accuracy drops noticeably with heavy frames.

■ Head Pose Range: Extreme head tilts (>30°) cause the polynomial model to extrapolate beyond the calibration convex hull, producing large errors at oblique gaze angles.

■ Fatigue: Extended eye-control sessions cause ciliary muscle fatigue. A rest-reminder and optional dwell-click mode (alternative to wink) are planned to reduce gesture effort during long sessions.

■ Platform Coverage: cv2.CAP_DSHOW is Windows-specific. A cross-platform capture backend selection is needed for Linux and macOS deployment.

THE PATH FORWARD

The most impactful near-term improvement is replacing the polynomial calibration with a small neural network (3-layer MLP) that learns a personalised non-linear gaze-to-screen mapping from the same calibration data — expected to reduce mean gaze error by 30–40%, particularly at screen corners where polynomial extrapolation is weakest.

A gaze typing integration — an on-screen keyboard where each key is selectable via dwell or wink — would make Neuro-Cursor a complete text-entry solution for non-verbal users. The existing PyAutoGUI foundation extends to a transparent overlay keyboard without changes to the core pipeline.

CONCLUSION

Neuro-Cursor demonstrates that professional-grade eye-controlled mouse interaction is achievable entirely in software, on commodity hardware, at zero additional cost. The combination of MediaPipe's accurate iris detection, a RANSAC-fitted degree-2 polynomial calibration, an Adaptive Kalman filter, and a wink-based gesture vocabulary produces a system that is both technically sound and practically usable.

The sixteen-point static + smooth-pursuit calibration pipeline, IQR outlier rejection, and post-calibration verification with pixel-level error reporting give Neuro-Cursor a level of rigour that matches — and in some respects exceeds — entry-level commercial alternatives costing tens of thousands of rupees.

For students and engineers in Computer Science, Neuro-Cursor is a demonstration of how classical signal processing (Kalman filtering, polynomial regression, RANSAC), modern deep-learning landmark detection (MediaPipe), and practical system engineering combine to solve a real human accessibility problem — without spending a single rupee on hardware.

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